r/Futurology

AI might be a net positive for the environment as of today. And more so in the future.

Posted by u/Reynvald • • 0 points • 15 comments

Please take all calculations and sources with a grain of salt, as such things are generally hard to quantify. I also would be happy to get corrected if I made mistakes or misrepresented some data. As well as your thoughts of the topic and how the environmental future could be look like in a 10-20-30 years.

I tried to balance myself out with sources from both sides. I'm an AI doomer myself, but at the same time I think that many today's environmental claims is an overreach. So I'm obviously have some biases.

IMO there are three main talking points about AI, harming the environment:

- Energy consumption
- Carbon footprint and Greenhouse gases (GHG) in general
- Water scarcity and pollution

I sorted sources about negative impacts at the top of each section and about negative impact - on the bottom.

1. Energy consumption

As of 2024, [Data centers accounted for about 1.5% of global electricity consumption, with AI accounted for 15% of total data centre energy demand accordingly.](https://www.scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030/) Therefore we can say that AI itself is using around 0.225% of global energy reserves.

[Predicted share of energy usage for data centers by 2030 is between 5 and 20%.](https://www.scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030/)
Considering that AI it still on it's growth and can take over up to 50% of all data center's resources, in 2030 it can be responsible for 2.5 up to 10% of all energy consumption (20 up to 90 times more, than of now) which is quite radical, but not unrealistic prediction.

Nevertheless, as of right now, [ML-related technologies is able to provide 15% improvement in grid efficiency and 10–20% increase in battery storage efficiency](https://www.mdpi.com/2073-4433/15/10/1250) and [20–30% relative efficiency gains in cell and module R&D.](https://journals.sagepub.com/doi/full/10.1177/0958305X241256293)
Same magnitude of efficiency gains is also the case for all clean and non-clean energy sources, by forecasting the weather and autoadjusting solar panels, micromanaging power grids and plants, predicting deposits of fossil energy sources and so on.

I think safe to say, that estimated energy gain overall will equal to or most likely surpass even the most pessimistic prognosis of 10% energy consumption from AI alone by 2030.

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2. Carbon footprint and GHG in general

According to [ICEF report from November 2024](https://www.icef.go.jp/wp-content/themes/icef/pdf/2024/roadmap/15_ICEF2.0%20GHG%20Emissions%20from%20AI_stand%20alone.pdf), (This link will download PDF file!) AI’s total GHG emissions are estimated at 100–300 million tonnes CO2, or roughly 0.2-0.6% of global emissions. With that, operational emissions are around 0.05% while manufacturing servers, chips, facilities, model trainings and life-cycle impacts make up the remainder.

At the same time AI can [reduce global GHG emissions by 5–10% by 2030](https://thesustainableagency.com/blog/environmental-impact-of-generative-ai/), via optimized grids, predictive maintenance, and smart agriculture and, additionally, [cuts of up to 5.3 gigatons CO2 (another 5–10% of current emissions)](https://www.sciencedirect.com/science/article/pii/S277242712200122X) - through applications in transport, buildings, and supply chains.

[One specific research](https://www.nature.com/articles/s41598-025-97110-3) (from month ago) from China indicates, that correlation between % of AI adoption and % of reducing carbon footprint (1% and 0.0395% accordingly) is quite sustainable and universal across the industries. But hard to say of this correlation will hold with future increasing AIfication of industries.

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3. Water scarcity and pollution

[Apparently in US AI is responsible for 0.5-0.7% of total annual water withdrawal](https://oecd.ai/en/wonk/how-much-water-does-ai-consume). If source took a data of water consumptions by data centers in general (it most likely the case, as most of the articles do so), then actual numbers will be a 15% of 0.5-0.7%, which is 0.075-0.105% accordingly.

Considering that most of the world AI infrastructure is located in US and China, safe to say, that for the rest of the world this percentages is significantly smaller.

The real concern, however, is the water pollution and separate cases of mismanagement from the corporations. [Quote: "Google’s planned data centre in Uruguay, which recently suffered its worst drought in 74 years, would require 7.6 million litres per day, sparking widespread protest."](https://sdgs.un.org/sites/default/files/2024-05/Gupta%2C%20et%20al._AIs%20excessive%20water%20consumption.pdf)
(This link will download PDF file!)

[Recent article from Politico](https://www.politico.com/news/2025/05/06/elon-musk-xai-memphis-gas-turbines-air-pollution-permits-00317582) about air pollution from xAI data center is also seems to me as a fair critique.

Here is a positive aspects:

[AI irrigation can reduce water usage by 30-50% while increasing yields by 20–30% (which is 5–8% savings of global agricultural withdrawals if deployed worldwide).](https://www.sciencedirect.com/science/article/pii/S2772375525002151)

[AI acoustic and pressure-based leak detection is already working and have 80–97% accuracy, cutting non-revenue water losses by 20–40%. Given that networks lose ~30% of supply globally (the most distant and arid places usually suffer the most), AI is saving 6–12% of treated water.
](https://www.waterrf.org/serve-file/2024_IWS-Challenge-Solution_Case-Western-CWD.pdf)
(This link will download PDF file!)

Same goes for demand forecasting, pump optimization, water quality assessment and many other projects, totaling up to 12% of the saved fresh water worldwide (if implemented worldwide as well, which is not the case for now). Some of this solutions is already implemented and working, although mostly in the water hungriest areas, like parts of Africa, China and India.

I think it's crucial to point out, that most of the water scarcity-related suffering is mostly occurring far from data centers and their water sources. And this problem is more of a logistical one (how to transport the water to the arid areas), than of sheer amount of fresh water world supplies.

I want to highlight, that AI still have an impact on environment and it's a right thing to strife for reducing the environmental impact in any area. But I believe that misinformation, toxicity and alarmism eventually will harm the both sides of this debates.

Comments

> **u/Such_Produce_7296** • 1 points • 2025-05-11 20:06 UTC
>
> Oh?
>
> so you used ai to sell that ai is not an electricity drain?
>
>> **u/Reynvald** • 1 points • 2025-05-11 20:16 UTC
>>
>> I used AI to speed up research on the topics, that I were previously overlooking. But I manually wrote the text and checked all the sources, if that's what you were saying.
>>
>> Basically, I wanted to organize my thoughts on that matter, by writing the post. And to hear other's opinions as a bonus.
>>
>> And, yea, I did kinda what you said. And ofc there is an electricity drain. But there is more to it, obviously.

> **u/Familiar_Gazelle_467** • 1 points • 2025-05-11 19:25 UTC
>
> TL:dr jevons paradox is a lie. Because OP thinks so
>
>> **u/Reynvald** • 1 points • 2025-05-11 19:40 UTC

Thanks for the jevons paradox. Didn't know the official name.

Although I agree with the premise, there is more to it.

For once, consuming more, however devilish it could sounds, is actually good. Not in terms of hamburgers or sodas, but in ability to increase quality of life by constructing more houses, provide easier access to water and education and so on. Jevons paradox (at least in your specific application) sounds like a consuming is an inherently bad thing. While it is surly a two sided blade for first world countries, even 10% Increase quality of life somewhere in rural Africa will not only raise people's appetite for good life, but also will reduce suffering and can help more people to realize their potential.

u/Familiar_Gazelle_467 • 1 points • 2025-05-11 20:40 UTC

Yw and I'm for all those good things of course. Raising the bar for quality of life is great it will just once again lead to more consumption, and a larger footprint on a finite earth.

u/hobopwnzor • 4 points • 2025-05-11 19:15 UTC

I don't see that you've included the extra energy to run the AI for all applications in the consideration.

If I build a new data center and buy solar panels and batteries for it I've made the percent of renewable data centers higher, but I haven't reduced emissions because I've just used more energy, and worse I've likely offset the ability for another data center doing work already to go green by taking the panels and space for AI.

u/Reynvald • 0 points • 2025-05-11 19:28 UTC

Hmm. I think the sources about AI, used in agriculture, power grids control, automation of other industries and so on is exactly about reducing emissions. There is also other AI-driven projects, like creating materials and 3d structures to catch emissions gases from the air, but I did't include thouse.

As for your last point — I'm not sure I understood you. Do you meant that one data center can cannibalize another due to limited amount of solar panels (and other green stuff) on the market? If so, I believe it's not a critical problem, because at least production of solar panels is actually quite scalable. No rare materials, simple construction. It even may have the opposite effect and stimulate the market of renewable energy.

u/hobopwnzor • 2 points • 2025-05-11 20:08 UTC

I think you should look more into everything that goes into building a renewable data center because it is a multiple year if not a decade-long process. They are very difficult to build and scale.

u/Reynvald • 1 points • 2025-05-11 20:21 UTC

Thanks for the tip! I will look into it. I sorta understand that it's not a simple process. I just initially thought that you meant a regular solar power plant that provide conventional energy to the data center. Not that power plant itself is a stones and sticks, but not a nuclear plant as well.

But yet again, I will look into it. Sounds like an interesting topic, actually.

u/BearsGotKhalilMack • 2 points • 2025-05-11 19:12 UTC

This was an interesting read, thank you for putting it all together. I don't think the data here is fully sufficient in proving that AI has a net environmental benefit in the present, sadly. The positive aspects that you listed are great, but you said it yourself: They're not implemented everywhere. Without specific evidence of how widespread their implementation is, I don't think it's quite enough to just assume that they're used ubiquitously enough to offset all of the well-documented energy/water issues.

Nonetheless, it is reassuring to know that there are positive environmental impacts of AI that may have the potential to eventually outweigh the very clear and widely-discussed negatives. If this really is the direction that the future of technology is headed, I do think our efforts are better spent trying to make it positive rather than just fighting it outright.

u/Reynvald • 1 points • 2025-05-11 19:16 UTC

Yes, you are correct to point it out and thanks for the critique.
In retrospect I also think that more of my positive arguments is more about the future, while negative ones - more

about today (although even it might look not so bad, compared with some wild claim in the darker corners of the internet :).

u/avalos-dario • 1 points • 2025-05-11 19:10 UTC

By destroying the planet faster, We Now have to fix it faster

u/Reynvald • 2 points • 2025-05-11 19:12 UTC

Sad, but true!

u/Fr00stee • 2 points • 2025-05-11 19:06 UTC

I think it can be a net positive as long as you use something like a nuclear reactor to power the data center instead of coal/gas

u/Reynvald • 1 points • 2025-05-11 19:08 UTC

For sure! Nuclear energy is undeservedly neglected. I'm all in on this.